

**SUNJOY AGRO**

NINGBO SUNJOY AGROSCIENCE CO., LTD.

Add: No.1165 Beihai Road, Chemical Industry Zone of Ningbo,
Zhenhai District, Ningbo, Zhejiang Province, China
Tel: 86-574-87774749 87760080
Fax: 86-574-87774751

MATERIAL SAFETY DATA SHEET

SECTION 1. CHEMICAL IDENTIFICATION

Product Name:	Difenoconazole 250 g/L EC
Company Name:	Ningbo Sunjoy Agrosience Co., Ltd.
Common Name of A.I.:	Difenoconazole
Chemical Name:	<i>cis,trans</i> -3-chloro-4-[4-methyl-2-(1 <i>H</i> -1,2,4-triazol-1-yl methyl)-1,3-dioxolan-2-yl]phenyl 4-chlorophenyl ether (IUPAC)
Synonym:	None
Chemical Family:	DMI: triazole
Chemical Formula:	C ₁₉ H ₁₇ Cl ₂ N ₃ O ₃
Usage:	Fungicide
Formulation:	Emulsifiable Concentrate

SECTION 2. COMPOSITION/INFORMATION ON INGREDIENTS

Component	CAS Number	Concentration (w/v)
Difenoconazole	119446-68-3	250 g/L
Other inert ingredient	/	Fill to 1L

Ingredients not precisely identified are proprietary or non-hazardous.

SECTION 3. HAZARDS IDENTIFICATION

Hazard classification

Hazardous. Irritating, dangerous for the environment.

Risk Phrases:



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R36 Irritating to eyes.

R50/53 Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

R65 Harmful : may cause lung damage if swallowed.

Safety phrases :

S2 Keep out of the reach of children.

S13 Keep away from food, drink and animal feedingstuffs.

S20/21 When using, do not eat, drink or smoke.

S36/37/39 Wear suitable protective clothing, gloves and eye/face protection.

S61 Avoid release to the environment. Refer to special instructions/Safety data sheets.

S62 If swallowed, do not induce vomiting : seek medical advice immediately and show this container or label.

SECTION 4. FIRST AID MEASURES

First Aid Measures

Ingestion: If swallowed: Call a poison control center or doctor immediately for treatment advice. Have the person sip a glass of water if able to swallow. Do not induce vomiting unless told to do so by a poison control center or doctor.

Eye Contact: If in eyes: Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after 5 minutes, then continue rinsing eye. Call a poison control center or doctor for treatment advice.

Skin Contact: If on skin or clothing: Take off contaminated clothing. Rinse skin immediately with plenty of water for 15-20 minutes. Call a poison control center or doctor for treatment advice.

Inhalation: If inhaled: Move person to fresh air. If person is not breathing, call an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a poison control center or doctor for further treatment advice.

Note to Physician



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There is no specific antidote. Treat symptomatically.

SECTION 5. FIRE FIGHTING MEASURES

Fire and Explosion

Flash Point: >61 °C

Auto-ignition Temperature: Not available

Flammability: Not flammable

Unusual Fire, Explosion and Reactivity Hazards

During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

In Case of Fire

Use dry chemical, foam or CO₂ extinguishing media. Wear full protective clothing and self-contained breathing apparatus. Evacuate nonessential personnel from the area to prevent human exposure to fire, smoke, fumes or products of combustion. Prevent use of contaminated buildings, area, and equipment until decontaminated. Water runoff can cause environmental damage. If water is used to fight fire, dike and collect runoff.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Emergency Procedures

In case of spillage it is important to take all steps necessary to:

- Avoid eye and skin contact.
- Avoid contamination of waterways

Methods and Materials for Containment and Clean Up

Procedure for spill:

- (1) Keep all bystanders away.
- (2) Wear goggles, half face-piece respirator with combined dust and vapour cartridge, full length clothing and PVC gloves.



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- (3) Reposition any leaking containers so as to minimise further leakage.
- (4) Dam and absorb spill with an absorbent material (e.g. sand or soil).
- (5) Shovel the absorbed spill into drums.
- (6) Disposal of the absorbed material will depend upon the extent of the spill.
 - For quantities up to 50L of product bury in a secure landfill site.
 - For quantities greater than 50L seek advice (use emergency contact number below) from the manufacturer before attempting disposal. Contain in a secure location until disposal method is established.
- (7) Decontaminate the spill area with detergent and water and rinse with water

SECTION 7. HANDLING AND STORAGE

Store the material in a well-ventilated, secure area out of reach of children and domestic animals. Do not store food, beverages or tobacco products in the storage area. Prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling.

SECTION 8. EXPOSURE CONTROLS/ PERSONAL PROTECTION

Exposure Standards

No exposure standards have been set for difenoconazole technical.

Engineering Controls

Provide adequate ventilation to keep airborne concentration below the exposure limit.

Personal Protection

Will irritate the eyes and skin. Avoid contact with eyes and skin. If product in eyes, wash it out immediately with water. When opening the container and preparing spray wear:

- cotton overalls buttoned to the neck and wrist and a washable hat,
- elbow-length PVC gloves and
- goggles



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- Self-contained breathing apparatus and total protection required in enclosed area

When using (except aerial spraying) the prepared spray, wear

- cotton overalls buttoned to the neck and wrist and a washable hat and
- elbow-length PVC gloves.

Wash hands after use. After each day's use, wash gloves, goggles and contaminated clothing.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance:	Amber clear liquid
Odor:	Non-specific
Melting Point:	82.0–83.0 °C (<i>difenoconazole</i>)
Boiling Point:	100.8 °C/3.7 mPa (<i>difenoconazole</i>)
Density:	1.02 – 1.05 (20°C)
pH:	5.0 to 8.0
Solubility in water:	Emulsifier in water
Vapor Pressure:	3.3×10^{-5} mPa (25 °C) (<i>difenoconazole</i>)

SECTION 10. STABILITY AND REACTIVITY

Stability:	Stable under normal use and storage.
Hazardous Polymerization:	Will not occur.
Conditions to Avoid:	Avoid high temperature.
Materials to Avoid:	Avoid strong oxidizer and alkaline
Hazardous Decomposition Products:	Will evolve toxic and irritant vapours.

SECTION 11. TOXICOLOGICAL INFORMATION

Acuter Toxicity/Irritation Studies (for the end product)

Acute Oral Toxicity Rat: LD50: >2000 mg/kg bw.

Acute Dermal Toxicity Rat: LD50: > 2000 mg/kg bw



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Acute Inhalation Toxicity	Rat: LC50 (4 h): > 5.63 mg/l
Eye Irritation	Rabbit: Irritating to eyes
Skin Irritation	Rabbit: Mildly irritating to skin
Sensitization	Guinea pig: A skin sensitizer

SECTION 12. ECOLOGICAL INFORMATION

Ecological Information (for difenoconazole technical):

Birds: Acute oral LD50 (9–11 d) for mallard ducks >2150, Japanese quail >2000 mg/kg b.w. Dietary LC50 (5 d) for bobwhite quail 4760, mallard ducks >5000 ppm.

Fish: LC50 (96 h) for rainbow trout 1.1, bluegill sunfish 1.2, sheepshead minnows 1.1 mg/l.

Daphnia: EC50 (48 h) 0.77 mg/l.

Algae: EC50 (72 h) for *Scenedesmus subspicatus* 0.03 mg/l.

Other aquatic spp.: LC50 (96 h) for mysid shrimps 0.15 mg/l; EC50 (96 h) for eastern oysters 0.3 mg/l; (7 d) for duckweed (*Lemna gibba*) 1.9 mg/l. NOEC (28 d) for development and emergence in the sediment dweller, *Chironomus riparius* 10 mg/kg sedimen and 0.4 mg/l.

Bees: Non-toxic to honeybees; LD50 (oral) >187 µg/bee; LC50 (contact) >100 µg/bee.

Worms: LC50 for earthworms >610 mg/kg dry soil.

Other beneficial spp.: NOEC (28 d) for reproduction in *Collembola* (*Folsomia candida*) 500 mg/kg soil. Use under field conditions is not expected to result in adverse effects on soil micro-organisms or on soil- and foliar-dwelling non-target arthropods.

Soil/Environment (for difenoconazole technical)

Practically immobile in soil, strong adsorption to soil particles (mean adsorption coefficient normalised to organic carbon, $K_{oc,ads}$ 4545 ml/g), low potential to leach below top soil layer. Soil dissipation rate is slow and dependent on application rate; DT50 3 mo–1 y. Hydrolytically stable at pH 5–9 (25 °C). Undergoes indirect photolysis in (sterile) natural water; DT50 4.6 d. In standard lab. water-sediment systems (n=2) in the dark, rapid dissipation from the water, DT50 1–3 d, but slow degradation in whole system, DT50 c. 8 mo.



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SECTION 13. DISPOSAL CONSIDERATIONS

Single rinse before disposal. Add rinsings to spray tank. DO NOT dispose of undiluted chemicals on site. Puncture or shred and bury empty containers in a local authority landfill. If no landfill is available, bury the containers below 500 mm in a disposal pit specifically marked and set up for this purpose clear of waterways, desirable vegetation and tree roots. Empty containers and product should not be burnt.

SECTION 14. TRANSPORT INFORMATION

Land transport ADR/RID

UN number: 3082

Proper Shipping Name: Environmentally Hazardous Substance, Liquid, n.o.s.
(contains difenoconazole)

Class: 9

Subsidiary Risk: Not regulated

Packaging group: III

Marine transport IMDG

UN number: 3082

Proper Shipping Name: Environmentally Hazardous Substance, Liquid, n.o.s.
(contains difenoconazole)

Class: 9

Subsidiary Risk: Not regulated

Packaging group: III

Marine pollutant: Yes

Air transport ICAO/IATA

UN number: 3082

Proper Shipping Name: Environmentally Hazardous Substance, Liquid, n.o.s.
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Class: 9



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Subsidiary Risk: Not regulated

Packaging group: III

SECTION 15. REGULATORY INFORMATION

EPCRA SARA Title III Classification:

Section 311/312 Hazard Classes: Acute Health Hazard

Section 313 Toxic Chemicals: Not applicable

California Proposition 65

Not applicable

CERCLA/SARA 302 Report Quantity (RQ)

None.

RCRA Hazardous Waste Classification (40 CFR 261)

Not applicable

TSCA Status

Exempt from TSCA, subject to FIFRA

SECTION 16. OTHER INFORMATION

Revised Date: December 17, 2015

MSDS Number: SN687/25%EC

Revision Summary:

To reflect database modification (no substantive changes - only related to system).

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